

CPPDescent

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Chapter 1

CPP-Descent

Part of the "Software Development for Computing Systems" course of the Department of Informatics and Telecommunications, UoA for the first semester of the academic year 2023-2024.

A library that implements the KNN-Graph creation and search algorithms described in [1], [2].

1.1 Dependencies

For local development, you will need Make and CMake. To install them on an apt-based linux distro, run the following commands:

```
$ sudo apt update && sudo apt upgrade
$ sudo apt install make
$ sudo apt install cmake
```

1.1.1 Minor dependencies

- The project uses Google Test for its testing needs. Gtest is used as a git submodule of this repo, that is downloaded by CMake the first time you clone. That means that nothing extra needs to be installed to run the tests.
- If you would like to generate the documentation, you will need doxygen installed.

1.2 Code Structure

In the wiki, you can find documentation generated by Doxygen. It is also available in pdf form.

1.3 Contributors

- Konstantinos Chousos (1115202000215)
- Anastasios-Fedon Seitanidis (1115202000179)

1.4 Bibliography

[1] W. Dong, C. Moses, and K. Li, “Efficient k-nearest neighbor graph construction for generic similarity measures,” in *Proceedings of the 20th international conference on World wide web*, in WWW '11. New York, NY, USA: Association for Computing Machinery, Mar. 2011, pp. 577–586. doi: [10.1145/1963405.1963487](https://doi.org/10.1145/1963405.1963487).

[2] “How PyNNDescent works — pynndescent 0.5.0 documentation.” Accessed: Oct. 10, 2023. [Online]. Available: https://pynndescent.readthedocs.io/en/latest/how_pynndescent_works.html

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 3

File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

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Chapter 4

Class Documentation

4.1 cppdescent Class Reference

A sample function for the cppdescent class, used to test the correctness of the CMake files.

```
#include <cppdescent.hpp>
```

Public Member Functions

- int [hello](#) (void)
Prints "Hello World".

4.1.1 Detailed Description

A sample function for the cppdescent class, used to test the correctness of the CMake files.

4.1.2 Member Function Documentation

4.1.2.1 hello()

```
int cppdescent::hello (  
    void )
```

Prints "Hello World".

Returns

int exit code

The documentation for this class was generated from the following files:

- [cppdescent.hpp](#)
- [cppdescent.cpp](#)

4.2 List Class Reference

ADT [List](#).

```
#include <ADTList.hpp>
```

Public Member Functions

- [List](#) (DestroyFunc destroyValue=nullptr)
Construct a new [List](#) object.
- [~List](#) ()
Destroy the [List](#) object.
- void [insertNext](#) ([ListNode](#) *node, Pointer value)
Adds a node after the given node.
- void [removeNext](#) ([ListNode](#) *node)
Removes the node after the given node.
- Pointer [find](#) (Pointer value, CompareFunc compare)
Finds the first value equal to the value parameter.
- DestroyFunc [setDestroyValue](#) (DestroyFunc value)
Set the Destroy Value object to a new value.
- int [getSize](#) ()
Get the size of the [List](#).
- int [increaseSize](#) ()
Increase the list size.
- int [decreaseSize](#) ()
Decrease the list size.
- [ListNode](#) * [getHead](#) ()
Getter for the head of the [List](#).
- [ListNode](#) * [getTail](#) ()
Getter for the tail of the [List](#).
- [ListNode](#) * [next](#) ([ListNode](#) *node)
Returns the node after the given one.
- Pointer [nodeValue](#) ([ListNode](#) *node)
Returns the value of the node.
- [ListNode](#) * [findNode](#) (Pointer value, CompareFunc compare)
Finds the first node that has value equal to the value parameter.

4.2.1 Detailed Description

ADT [List](#).

An Abstract Data Type [List](#) with linear access to data, and addition and removal to/from arbitrary positions.

4.2.2 Constructor & Destructor Documentation

4.2.2.1 List()

```
List::List (
    DestroyFunc destroyValue = nullptr )
```

Construct a new [List](#) object.

Parameters

<i>destroyValue</i>	If <i>destroyValue</i> != nullptr, the <i>destroyValue</i> function will be called each time a node is removed.
---------------------	---

head is a virtual node that is always present in the list.

Parameters

<i>destroyValue</i>	The destroy function for the values of the list. Defaults to nullptr.
---------------------	---

4.2.2.2 ~List()

```
List::~~List ( )
```

Destroy the [List](#) object.

If the *destroyValue* function is not nullptr, it is called for every value.

4.2.3 Member Function Documentation**4.2.3.1 decreaseSize()**

```
int List::decreaseSize ( )
```

Decrease the list size.

Returns

int The updated size.

4.2.3.2 find()

```
Pointer List::find (
    Pointer value,
    CompareFunc compare )
```

Finds the first value equal to the value parameter.

Find the first value equal to the value parameter.

Parameters

<i>value</i>	The value based on which the search is done.
<i>compare</i>	A pointer to the function that will compare the node values.

Returns

Pointer A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

Pointer A pointer to found value.

4.2.3.3 findNode()

```
ListNode * List::findNode (
    Pointer value,
    CompareFunc compare )
```

Finds the first node that has value equal to the value parameter.

Find the first node with value equal to the value parameter.

Parameters

<i>value</i>	The value to search.
<i>compare</i>	The compare function to use.

Returns

ListNode* A pointer to the resulting node.

Parameters

<i>value</i>	The value to search for.
<i>compare</i>	A pointer to the compare function.

Returns

ListNode* A pointer to the found node.

4.2.3.4 getHead()

```
ListNode * List::getHead ( )
```

Getter for the head of the [List](#).

Get the first element of the [List](#).

Returns

ListNode* The head.

Returns

ListNode* Pointer to the first element.

4.2.3.5 getSize()

```
int List::getSize ( )
```

Get the size of the [List](#).

Get the size of the list.

The size is updated upon entry/removal of nodes, so the retrieval of the size is O(1).

Returns

int

Returns

int The size.

4.2.3.6 getTail()

```
ListNode * List::getTail ( )
```

Getter for the tail of the [List](#).

Get the last element of the list.

Returns

ListNode* The tail.

If the list is empty, the virtual head is the tail, so we return nullptr.

Returns

ListNode* Pointer to the last element.

4.2.3.7 increaseSize()

```
int List::increaseSize ( )
```

Increase the list size.

Returns

int The updated size.

4.2.3.8 insertNext()

```
void List::insertNext (
    ListNode * node,
    Pointer value )
```

Adds a node *after* the given node.

Insert a new element after the node.

Parameters

<i>node</i>	The node that the new node will be placed after.
<i>value</i>	The value of the new node.

Parameters

<i>node</i>	The node after which the element is inserted.
<i>value</i>	The value of the new element.

4.2.3.9 next()

```
ListNode * List::next (
    ListNode * node )
```

Returns the node after the given one.

Get the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode* A pointer to the next node.

Parameters

<i>node</i>	
-------------	--

Returns

ListNode*

4.2.3.10 nodeValue()

```
Pointer List::nodeValue (  
    ListNode * node )
```

Returns the value of the node.

Get the node's value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer A generic pointer to the value.

Parameters

<i>node</i>	
-------------	--

Returns

Pointer

4.2.3.11 removeNext()

```
void List::removeNext (  
    ListNode * node )
```

Removes the node *after* the given node.

Removes the node after the given node.

Parameters

<i>node</i>	The node before the node to be removed.
-------------	---

Parameters

<i>node</i>	The node after which the element is removed.
-------------	--

4.2.3.12 setDestroyValue()

```
DestroyFunc List::setDestroyValue (
    DestroyFunc value )
```

Set the Destroy Value object to a new value.

Set the destroy value function.

Parameters

<i>value</i>	A pointer to the new destroy function.
--------------	--

Returns

DestroyFunc A pointer to the old destroy function.

Parameters

<i>value</i>	The new destroy function.
--------------	---------------------------

Returns

DestroyFunc The old destroy function.

The documentation for this class was generated from the following files:

- ADTList.hpp
- [ADTList.cpp](#)

4.3 ListNode Class Reference

Node of a list.

```
#include <ADTList.hpp>
```

Public Member Functions

- [ListNode](#) ()
Construct a new [List](#) Node object that is empty.
- [ListNode](#) (Pointer value)
Construct a new [List](#) Node object with value 'value'.
- void [setNext](#) ([ListNode](#) *next)
Destroy the [List](#) Node object.
- [ListNode](#) * [getNext](#) ()
Getter for the next node.
- Pointer [getValue](#) ()
Getter for the value of the node.

4.3.1 Detailed Description

Node of a list.

A list node is represented by its value. It also holds a pointer to the next node in the list.

4.3.2 Constructor & Destructor Documentation

4.3.2.1 [ListNode\(\)](#) [1/2]

```
ListNode::ListNode ( ) [inline]
```

Construct a new [List](#) Node object that is empty.

4.3.2.2 [ListNode\(\)](#) [2/2]

```
ListNode::ListNode (
    Pointer value ) [inline]
```

Construct a new [List](#) Node object with value 'value'.

Parameters

<i>value</i>	a generic pointer to the value of the node.
--------------	---

4.3.3 Member Function Documentation

4.3.3.1 getNext()

```
ListNode * ListNode::getNext ( )
```

Getter for the next node.

Returns

ListNode* The pointer to the next node.

4.3.3.2 getValue()

```
Pointer ListNode::getValue ( )
```

Getter for the value of the node.

Returns

Pointer A generic pointer to the value.

4.3.3.3 setNext()

```
void ListNode::setNext (
    ListNode * next )
```

Destroy the [List](#) Node object.

Setter for the next node.

Parameters

<i>next</i>	The pointer to the next node.
-------------	-------------------------------

The documentation for this class was generated from the following files:

- ADTList.hpp
- [listNode.cpp](#)

Chapter 5

File Documentation

5.1 ADTList.hpp

```
00001
00011 #pragma once
00012 #include "common.hpp"
00013
00014 #define LIST_BOF (ListNode*)0
00015 #define LIST_EOF (ListNode*)0
00016
00024 class ListNode {
00025 public:
00030     ListNode() : next(nullptr), value(nullptr){};
00036     ListNode(Pointer value) : next(nullptr), value(value){};
00041     // ~ListNode();
00047     void setNext(ListNode* next);
00053     ListNode* getNext();
00059     Pointer getValue();
00060
00061 private:
00062     ListNode* next;
00063     Pointer value;
00064 };
00065
00073 class List {
00074 public:
00081     List(DestroyFunc destroyValue = nullptr);
00086     ~List();
00093     void insertNext(ListNode* node, Pointer value);
00099     void removeNext(ListNode* node);
00107     Pointer find(Pointer value, CompareFunc compare);
00114     DestroyFunc setDestroyValue(DestroyFunc value);
00123     int getSize();
00124     int increaseSize();
00125     int decreaseSize();
00131     ListNode* getHead();
00137     ListNode* getTail();
00144     ListNode* next(ListNode* node);
00151     Pointer nodeValue(ListNode* node);
00159     ListNode* findNode(Pointer value, CompareFunc compare);
00160
00161 private:
00162     ListNode* head;
00163     ListNode* tail;
00164     int size;
00165     DestroyFunc destroyValue;
00166 };
```

5.2 common.hpp

```
00001 #pragma once
00002
00003 typedef void* Pointer;
00004
00005 typedef int (*CompareFunc)(Pointer a, Pointer b);
00006
00007 typedef void (*DestroyFunc)(Pointer value);
```

5.3 cppdescent.hpp

```
00001
00006 class cppdescent {
00007 public:
00013     int hello(void);
00014 };
```

5.4 ADTList.cpp File Reference

Implementation of an ADT linked list.

```
#include "cppdescent/ADTList.hpp"
Include dependency graph for ADTList.cpp:
```

5.5 cppdescent.cpp File Reference

Implementation of cppdescent library.

```
#include "cppdescent/cppdescent.hpp"
#include <iostream>
Include dependency graph for cppdescent.cpp:
```

5.5.1 Detailed Description

Implementation of cppdescent library.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-27

Copyright

Copyright (c) 2023

5.6 listNode.cpp File Reference

Implementation of the listNode to be used with linkedList.

```
#include "cppdescent/ADTList.hpp"
Include dependency graph for listNode.cpp:
```

5.6.1 Detailed Description

Implementation of the listNode to be used with linkedList.

Author

Konstantinos Chousos

Version

0.1

Date

2023-10-29

Copyright

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